ALCOHOL DETECTION SYSTEM

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1. Problem Statement

1.1 Introduction:

Road safety is a major concern worldwide, with a significant number of traffic accidents caused by drivers operating vehicles under the influence of alcohol. Alcohol consumption impairs judgment, slows reaction time, and reduces a driver's ability to make safe decisions, increasing the likelihood of collisions and fatalities. Although governments enforce strict laws against drunk driving, violations continue to occur due to the absence of continuous monitoring before or during vehicle operation.

1.2 Problem Definition:

Despite strict regulations and awareness campaigns, drunk driving remains one of the leading causes of preventable road accidents. Conventional enforcement methods, such as roadside breathalyzer tests, depend on manual intervention by law enforcement agencies and cannot continuously monitor every vehicle. As a result, many intoxicated drivers can operate vehicles without immediate detection.

The challenge is to develop an **embedded system** that automatically detects alcohol presence before or during vehicle operation and provides an immediate response. The system should be affordable, compact, easy to install, and capable of providing reliable alcohol detection without requiring complex infrastructure. In this project, the response is demonstrated through visual indicators and an audible alarm.

1.3 Literature and Market Review:

Several technologies have been proposed and implemented to reduce accidents caused by alcohol-impaired driving. The most common approaches include handheld breath analyzers, alcohol ignition interlock devices, embedded alcohol detection systems, and Internet of Things (IoT)-based monitoring systems. Each approach differs in terms of cost, accuracy, automation, and ease of implementation.

1.4 Existing Solutions:

1. Handheld Breath Analyzers:

Handheld breath analyzers are widely used by law enforcement agencies to estimate a driver's breath alcohol concentration. These devices provide accurate measurements but require manual operation and cannot continuously monitor the driver's condition during vehicle operation. They are primarily intended for roadside enforcement rather than integration into vehicles.

2. Alcohol Ignition Interlock Devices (AIIDs):

Alcohol ignition interlock systems require the driver to provide a breath sample before the vehicle can be started. If alcohol above a predefined threshold is detected, the ignition system remains disabled. These systems have demonstrated high effectiveness in preventing repeat drunk-driving incidents but involve relatively high installation and maintenance costs, limiting their widespread adoption.

3. IoT-Based Alcohol Monitoring Systems:

Modern research extends alcohol detection systems by integrating wireless communication technologies such as ESP8266 Wi-Fi modules. These systems transmit alcohol detection data to cloud platforms or mobile applications for remote monitoring and logging. While these solutions provide enhanced functionality, they increase system complexity, cost, and power consumption.

2. Scope of the Solution

2.1 Objectives:

The primary objective of this project is to design and develop a low-cost embedded alcohol detection system capable of identifying the presence of alcohol vapors using an MQ-3 alcohol sensor and an Arduino Uno microcontroller. Using this device, we are able to prevent drunk driving and save lives.

The project also aims to:

- Develop a reliable prototype using readily available electronic components.
- Demonstrate real-time alcohol detection through visual and audible alerts.
- Simulate the concept of an engine-locking mechanism that can be integrated into future vehicle safety systems.
- Design a custom PCB and generate Gerber files for manufacturing.
- Validate the circuit through simulation and hardware implementation.

2.2 Scope and Boundaries:

The proposed solution is intended as a **prototype** demonstrating the feasibility of alcohol detection in an embedded system. The project focuses on detecting alcohol vapors near the driver and generating appropriate alerts based on sensor readings.

The scope of this project includes:

- Alcohol detection using the MQ-3 gas sensor.
- Signal processing using the Arduino Uno and Arduino IDE for code implementation.
- Visual indication using red and green LEDs.
- Audible warning using a buzzer.
- Circuit simulation in Tinkercad.
- PCB schematic, layout, and Gerber file generation using EasyEDA.
- Hardware demonstration of the prototype.

The project does not include direct integration with a vehicle's ignition system or commercial vehicle electronics, as its primary objective is to demonstrate the detection and alert mechanism.

2.3 Limitations:

Although the proposed system effectively demonstrates alcohol detection, it has certain limitations:

- The MQ-3 sensor detects alcohol vapors rather than directly measuring Blood Alcohol Concentration (BAC).
- Sensor readings may be influenced by environmental conditions, temperature, humidity, and the presence of other volatile substances.
- The sensor requires a warm-up period and periodic calibration to maintain measurement accuracy.
- The engine-locking function is demonstrated conceptually and is not physically connected to an actual vehicle ignition system.
- The prototype is designed for educational and proof-of-concept purposes rather than commercial deployment.

These limitations are primarily due to the project's emphasis on affordability, simplicity, and the use of readily available components within the available development time.

2.4 Justification of the Proposed Solution:

The proposed solution offers a practical and economical approach to demonstrating alcohol detection using embedded systems. Compared with commercial ignition interlock systems, the prototype requires significantly fewer components while still illustrating the core principles of alcohol sensing, signal processing, and user alert generation.

The use of the Arduino Uno simplifies system development and programming, while the MQ-3 sensor provides an inexpensive and widely adopted method for detecting alcohol vapors. The inclusion of LEDs and a buzzer enables immediate feedback, making the system easy to understand and operate.

Furthermore, the project demonstrates the complete hardware development workflow by progressing from circuit simulation to PCB design and Gerber file generation. This makes the solution suitable not only as a proof-of-concept for vehicle safety applications but also as an educational platform for learning embedded systems, PCB design, and electronic prototyping.

Overall, the proposed system achieves a balance between functionality, cost, simplicity, and scalability, providing a strong foundation for future enhancements such as engine locking, GPS tracking, wireless communication, and cloud-based monitoring.

3. Block Diagram:

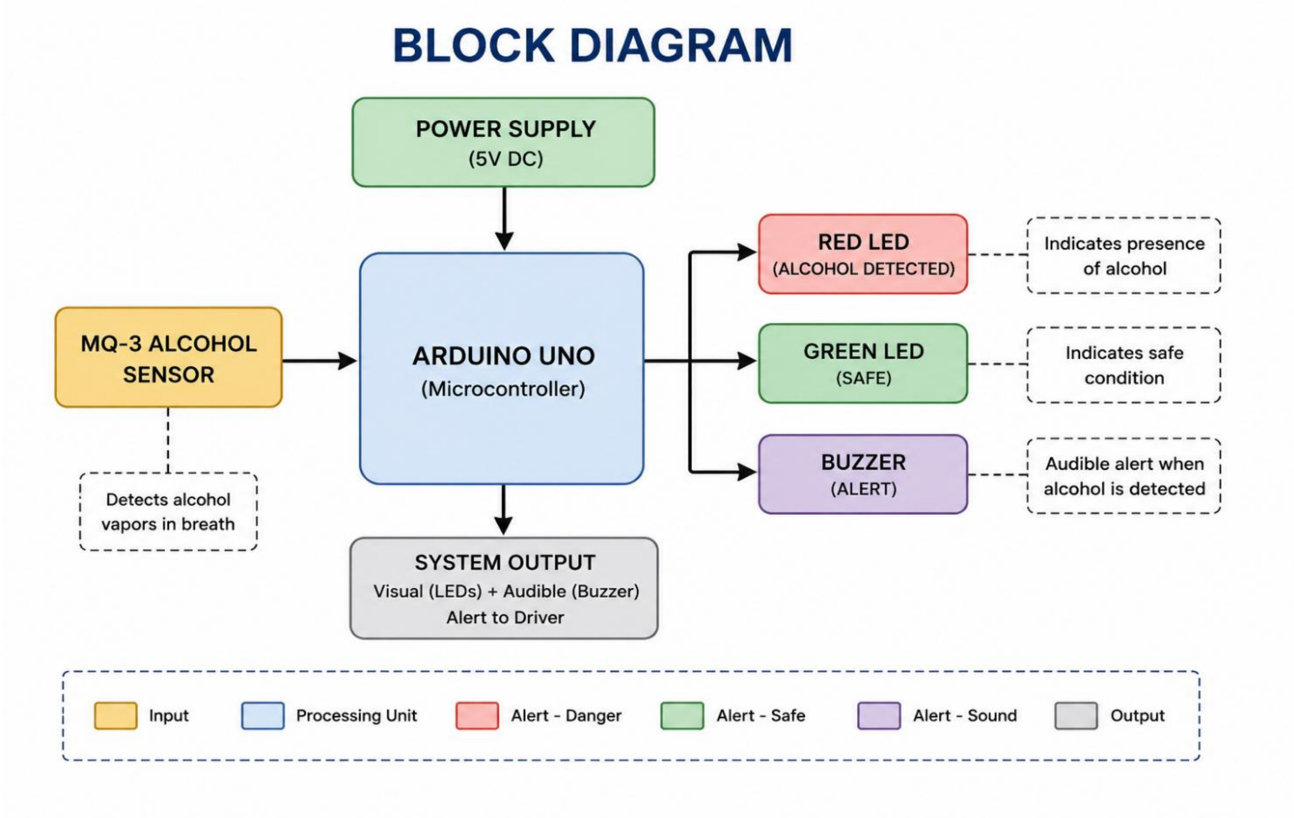


Fig 3.1:The figure gives a detailed overview of the project in the form of block diagram

4. Circuit Diagram:

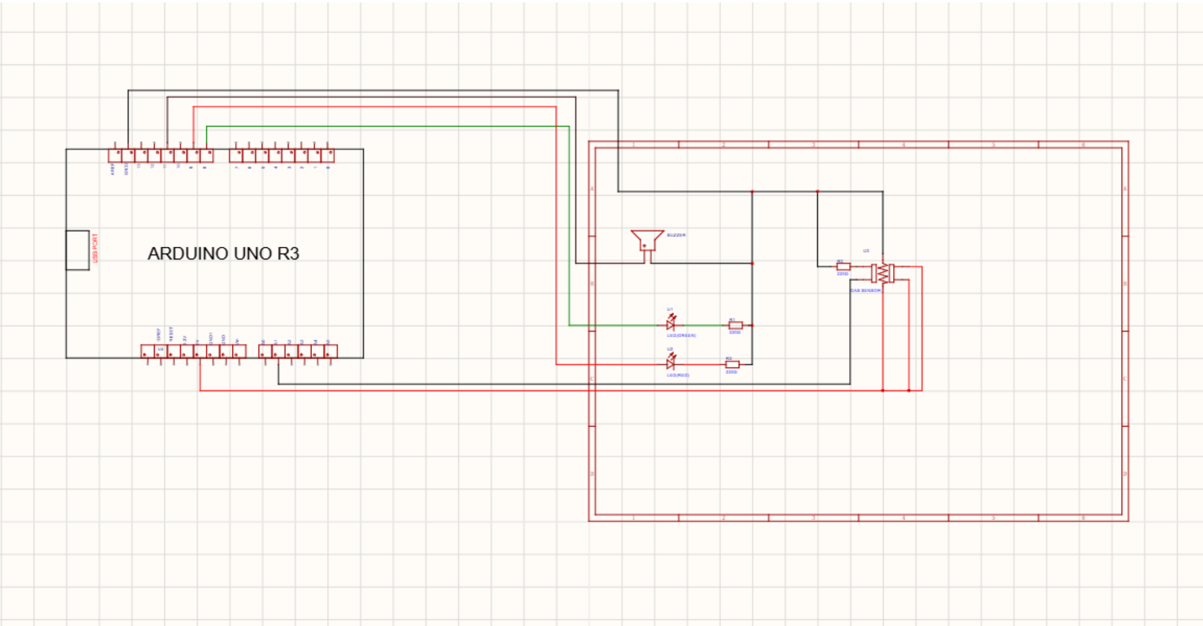
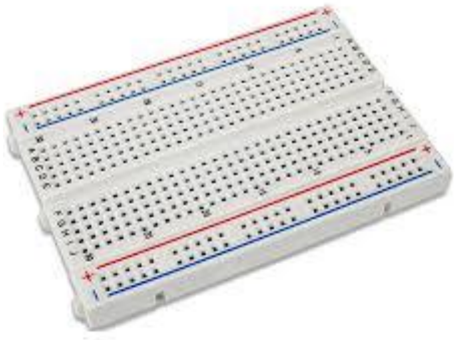


Fig 4.1:This circuit diagram is designed using the EasyEDA software

5. Components Required



1.ARDUINO UNO R3-(1)

2.BREADBOARD-(1)

3. LED(s)-(2)

4. 220Ω RESISTORS-(2)

5. BUZZER-(1)

5. MQ-3 SENSOR-(1)

6. JUMPER WIRES-(5)

6. Simulated Circuit

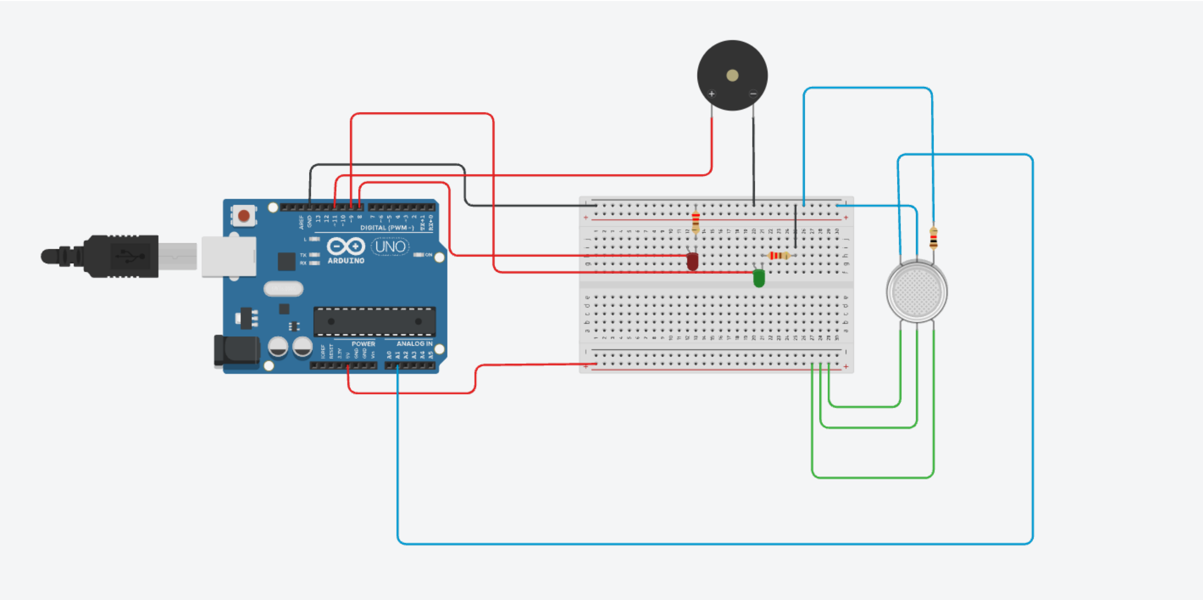


Fig 6.1:This Circuit is designed using Tinkercad

Circuit Functionality (Simulated Circuit)

The alcohol detection system was first designed and tested in Tinkercad before hardware implementation. Since the MQ-3 alcohol sensor is not available in the Tinkercad component library, a Gas Sensor was used in the simulation. This gas sensor behaves similarly, producing an analog output voltage that varies with gas concentration, allowing the logic of the alcohol detection system to be verified.

Working Principle

1. **Gas Sensor (Simulation Input):**
   The gas sensor continuously senses the surrounding environment and produces an analog output corresponding to the detected gas concentration. In the simulation, this sensor represents the MQ-3 alcohol sensor used in the actual hardware implementation. It has 6 pins with h1&h2 as heater pins connected to the 5V pin and GND, then pins A1&A2 are connected to the 5V terminal, B1 is connected to GND, and B2 is connected to the analog pin A1.

2. **Arduino Uno (Processing Unit):**
   The analog output of the gas sensor is connected to the A1 (Analog Pin 1) of the Arduino Uno. The Arduino continuously reads the sensor value using its Analog-to-Digital Converter (ADC) and compares it with a predefined threshold. The board uses **Embedded C** programming to control the external components.

3. **<u>Safe Condition (No Alcohol Detected):</u>**
   When the sensor value remains below the threshold, the Arduino considers the environment to be safe. It turns ON the Green LED while keeping the Red LED and Buzzer OFF. The green LED serves as an indication that no alcohol has been detected.

4. **<u>Alcohol Detected (Danger Condition):</u>**
   When the sensor value exceeds the predefined threshold, the Arduino interprets this as the presence of alcohol vapors. It immediately turns OFF the Green LED, ON the Red LED, and activates the Buzzer to provide both visual and audible warnings.

5. **<u>Threshold-Based Decision Making:</u>**
   The threshold value acts as the decision point for the system. Sensor readings below the threshold indicate safe conditions, whereas readings above the threshold trigger the warning system. A small delay and hysteresis are incorporated in the program to prevent rapid switching caused by fluctuating sensor values.

6. **<u>Power Supply:</u>**
   The entire circuit is powered by the 5 V supply from the Arduino Uno. The gas sensor, LEDs, and buzzer receive power from the Arduino, while all components share a common ground connection.

<u>Note:</u>

The Gas Sensor used in the Tinkercad simulation is only a substitute for demonstration purposes because the MQ-3 alcohol sensor is not available in the simulator. During actual hardware implementation, the Gas Sensor is replaced with the MQ-3 alcohol sensor, while the circuit connections and program logic remain the same.

7. Code for the Solution

```
// Alcohol Detection System
//Embedded C programming

const int sensorPin = 2;  // MQ-3 sensor
const int redLED = 8;       // Danger red LED
const int greenLED = 9;     // Safe  green LED
const int buzzer = 11;       // Buzzer


void setup()
{
   pinMode(sensorPin, INPUT);
   pinMode(redLED, OUTPUT);
   pinMode(greenLED, OUTPUT);
   pinMode(buzzer, OUTPUT);

   Serial.begin(9600);



}

void loop()
{
   int sensorState = digitalRead(sensorPin); //Intake of the input from MQ-3 sensor

//Alcohol detected

   if(sensorState == LOW)
   {
      digitalWrite(redLED, HIGH);
      digitalWrite(greenLED, LOW);
```

```
        digitalWrite(buzzer, HIGH);

        Serial.println("ALCOHOL DETECTED");
        Serial.println("NOT SAFE TO DRIVE");
    }

 //Alcohol not detected
    else
    {
        digitalWrite(redLED, LOW);
        digitalWrite(greenLED, HIGH);
        digitalWrite(buzzer, LOW);

        Serial.println("SAFE TO DRIVE");
    }

    delay(500);
}
```

8. Results of the Simulation

The alcohol detection system was successfully simulated using **Tinkercad** to verify the functionality of the designed circuit before hardware implementation. Since the MQ-3 alcohol sensor is unavailable in Tinkercad, a Gas Sensor was used to simulate alcohol detection. The simulation confirmed that the circuit operated as intended under different sensor input conditions.

Observations:

Case 1: Safe Condition (No Alcohol Detected)

- The Gas Sensor output remained below the predefined threshold.
- The Arduino Uno identified the environment as safe.
- The **Green LED** turned ON.
- The **Red LED** remained OFF.
- The **Buzzer** remained OFF.
- The Serial Monitor displayed sensor values below the threshold.

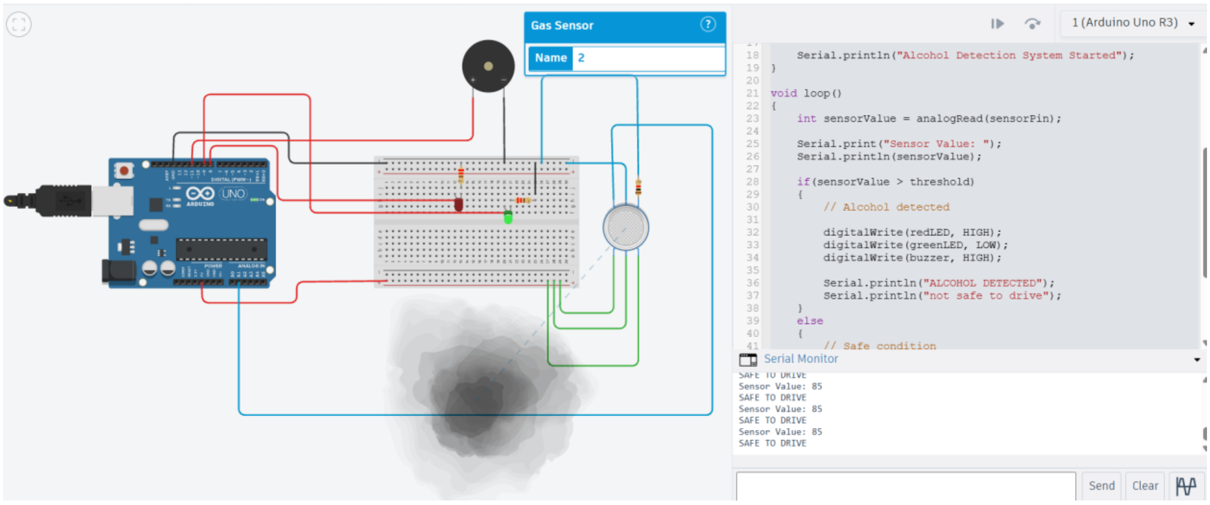


Fig 8.1:This figure successfully demonstrates the safe condition of the system using simulation

Case 2: Alcohol Detected

- When the Gas Sensor output exceeded the threshold, the Arduino detected the presence of alcohol.
- The **Green LED** turned OFF.
- The **Red LED** turned ON.
- The **Buzzer** was activated, providing an audible warning.
- The Serial Monitor displayed sensor values above the threshold.

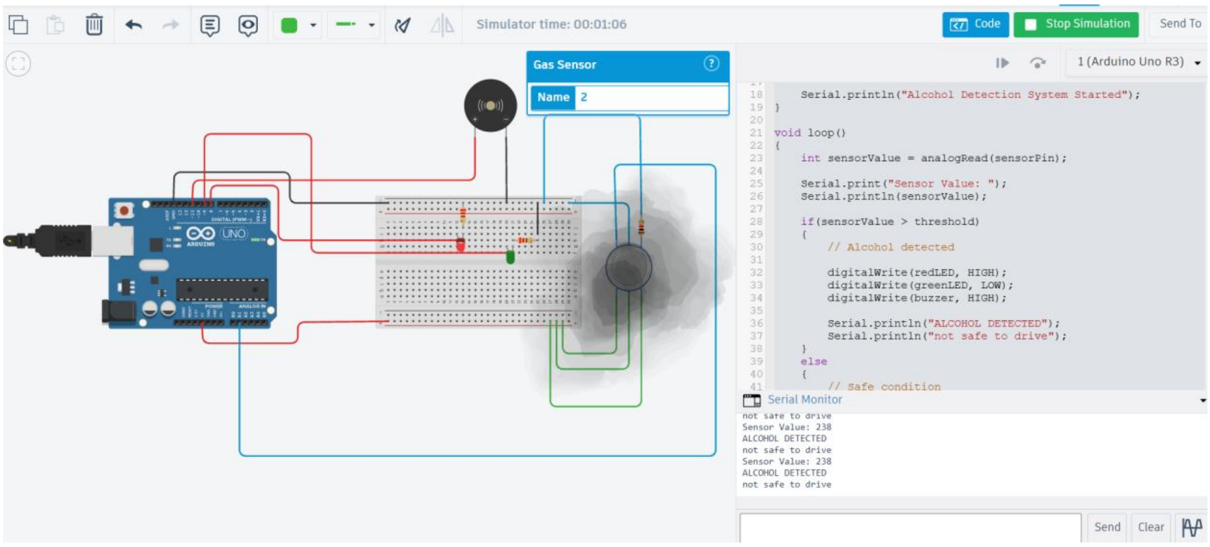


Fig 8.2:This figure successfully demonstrates the non-safe condition of the system using simulation

Simulation Outcome:

The simulation verified that the proposed alcohol detection system correctly monitored the gas sensor output, processed the analog signal using the Arduino Uno, and generated appropriate visual and audible alerts based on the detected gas concentration. The successful simulation confirmed that the circuit logic and program were functioning correctly and were suitable for hardware implementation using the MQ-3 alcohol sensor.

9. Layout of the System

The system layout represents the physical arrangement of all electronic components on the printed circuit board (PCB). The PCB was designed using **EasyEDA** to provide a compact, organized, and reliable implementation of the alcohol detection system.

The Arduino Uno is connected externally through header pins provided on the PCB, while the MQ-3 alcohol sensor, LEDs, buzzer, and current-limiting resistors are mounted on the board. The components are positioned to minimize wiring complexity and facilitate efficient routing of copper traces. The PCB layout also ensures proper power distribution by connecting all components to the common 5 V and GND lines.

The copper traces replace conventional jumper wires used during breadboard prototyping, resulting in improved mechanical stability and electrical reliability. The completed PCB layout serves as the final hardware design and was subsequently used to generate Gerber files for PCB manufacturing.

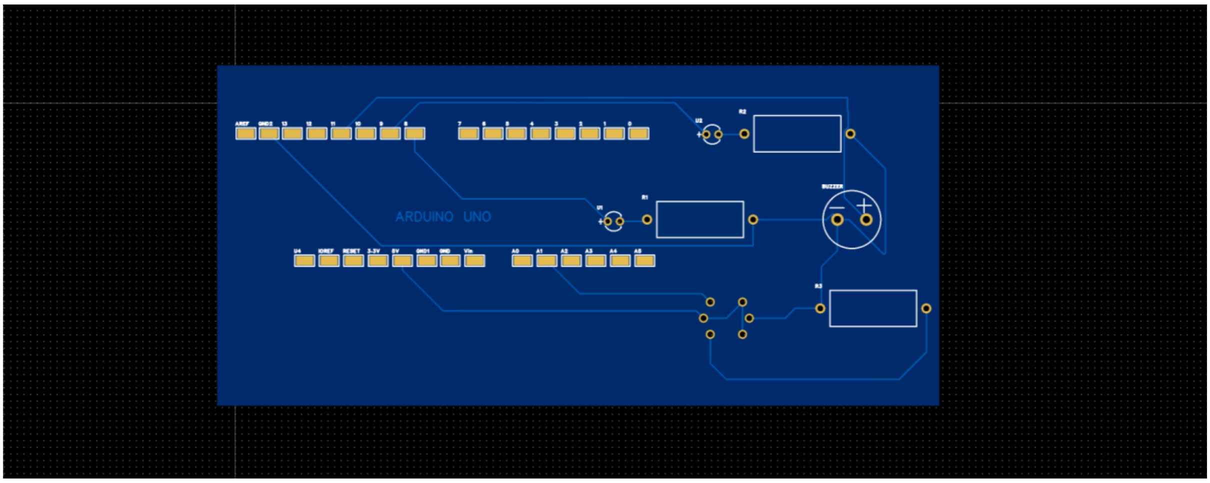


Fig 9.1:2D PCB layout of the proposed alcohol detection system designed using EasyEDA.

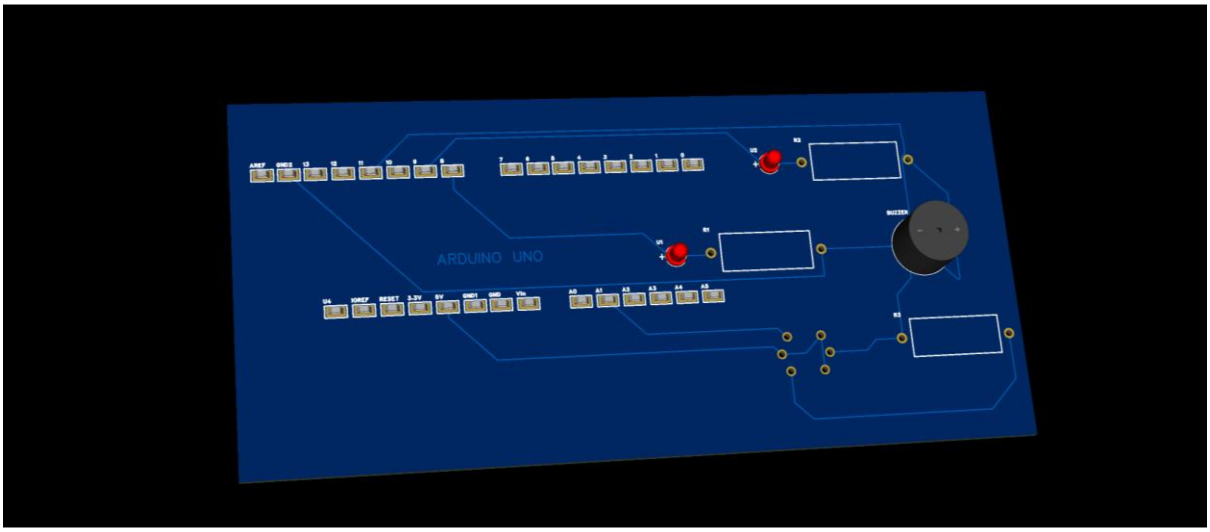
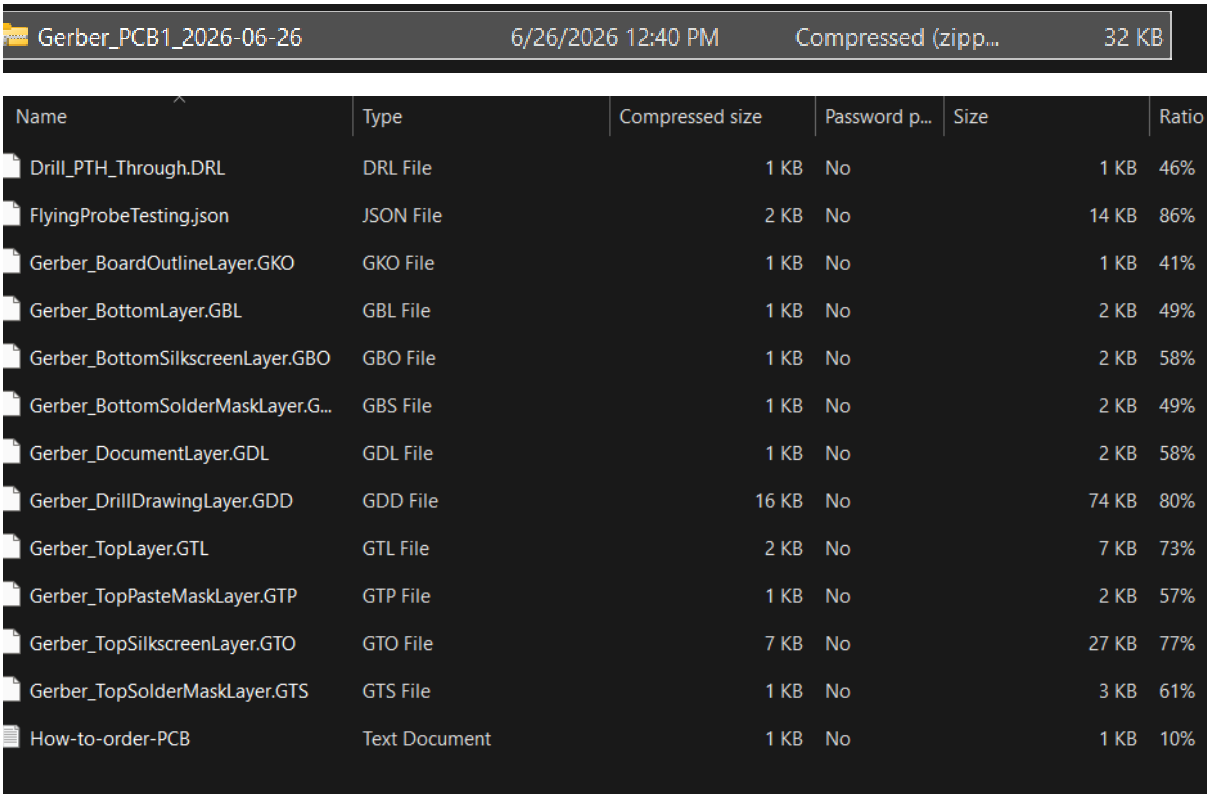


Fig 9.2:3D visualization of the designed PCB using EasyEDA.

10. Gerber File:

After completing the PCB layout and verifying the design rules, Gerber files were generated using **EasyEDA**. The Gerber package contains all the manufacturing data required for PCB fabrication, including copper layers, solder mask, silkscreen, drill files, and board outline. The generated files were exported as a compressed ZIP archive, which can be directly used by PCB manufacturers. The Gerber file can be found in the provided GitHub link.



Gerber_PCB1_2026-06-26	6/26/2026 12:40 PM	Compressed (zipp...	32 KB

Name	Type	Compressed size	Password p...	Size	Ratio
Drill_PTH_Through.DRL	DRL File	1 KB	No	1 KB	46%
FlyingProbeTesting.json	JSON File	2 KB	No	14 KB	86%
Gerber_BoardOutlineLayer.GKO	GKO File	1 KB	No	1 KB	41%
Gerber_BottomLayer.GBL	GBL File	1 KB	No	2 KB	49%
Gerber_BottomSilkscreenLayer.GBO	GBO File	1 KB	No	2 KB	58%
Gerber_BottomSolderMaskLayer.G...	GBS File	1 KB	No	2 KB	49%
Gerber_DocumentLayer.GDL	GDL File	1 KB	No	2 KB	58%
Gerber_DrillDrawingLayer.GDD	GDD File	16 KB	No	74 KB	80%
Gerber_TopLayer.GTL	GTL File	2 KB	No	7 KB	73%
Gerber_TopPasteMaskLayer.GTP	GTP File	1 KB	No	2 KB	57%
Gerber_TopSilkscreenLayer.GTO	GTO File	7 KB	No	27 KB	77%
Gerber_TopSolderMaskLayer.GTS	GTS File	1 KB	No	3 KB	61%
How-to-order-PCB	Text Document	1 KB	No	1 KB	10%

Fig 10.1:Proof of successful creation of Gerber files for the PCB design

11. Demo of the Video:

The demo of the project is posted in the provided GitHub link:

https://github.com/quasar2811-cpu/Alcohol_detection_system.git

12. References:

Research Papers

[1] M. Akhil, P. R. Praveen, K. S. Sreekanth, and M. S. Nair, "Alcohol Detection and Engine Locking System," *International Journal of Innovation Studies (IJIS)*, vol. 3, no. 2, 2025.

[2] S. Ram Kumar, R. Pradeep, and S. Karthik, "A Study on the Efficacy of Alcohol Detection in the Prevention of Drunken Driving," *IOP Conference Series: Materials Science and Engineering*, vol. 1145, 2021.

[3] K. Ravi Kumar, M. Harsha Vardhan, and P. Sai Teja, "Review Paper on Alcohol Detection and Vehicle Engine Locking System," *Journal of Pharmaceutical Negative Results*, 2022.

[4] Munasir and A. I. Ikhsan, "Design of an Alcohol Detector Using MQ-3 Sensor Based on Arduino Nano V3," *Inovasi Fisika Indonesia*, vol. 11, no. 3, pp. 81–87, 2022.

Datasheets

[5] Hanwei Electronics, *MQ-3 Alcohol Gas Sensor Datasheet*.

[6] Arduino, *Arduino Uno Rev3 Technical Specifications and Datasheet*.

Official Documentation

[7] Arduino Documentation – Programming Reference and Arduino IDE Documentation.

[8] EasyEDA Documentation – PCB Design, Schematic Capture, PCB Layout, and Gerber File Generation.

[9] Tinkercad Circuits Documentation – Circuit Simulation and Arduino Programming Environment.

Software and IDE Used

- Arduino IDE (Programming and Code Upload)
- Tinkercad Circuits (Circuit Simulation)
- EasyEDA Pro (Schematic Design, PCB Layout, Gerber Generation)
- Microsoft Word (Documentation)
- GitHub (Project Repository and Version Control)

Online Resources

- Arduino Official Documentation
- EasyEDA Official Documentation
- Tinkercad Official Documentation
- MQ-3 Sensor Datasheet

- World Health Organization (WHO) – Road Safety Reports
- National Highway Traffic Safety Administration (NHTSA) – Alcohol-Impaired Driving Statistics

GitHub Link:

https://github.com/quasar2811-cpu/Alcohol_detection_system.git

Tools Used

Category	Tool
Programming IDE	Arduino IDE
Simulation	Tinkercad Circuits
PCB Design	EasyEDA Pro
Programming Language	Embedded C / Arduino C++
Version Control	GitHub
Microcontroller	Arduino Uno R3
Sensor	MQ-3 Alcohol Sensor